

Annex 3: Arbitrary Baud Rate Calculation and Configuration

Instruction Manual

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1. Use of the Baud Rate Calculation Tool

In order to easily calculate the baud rate of our USBCAN adapter/CANalyst-II analyzer, you can use the tool software shown in Figure 1. (Placed in the USB_CAN TOOL software installation directory C:\Program Files (x86)\USB_CAN TOOL\BtrCal.exe)

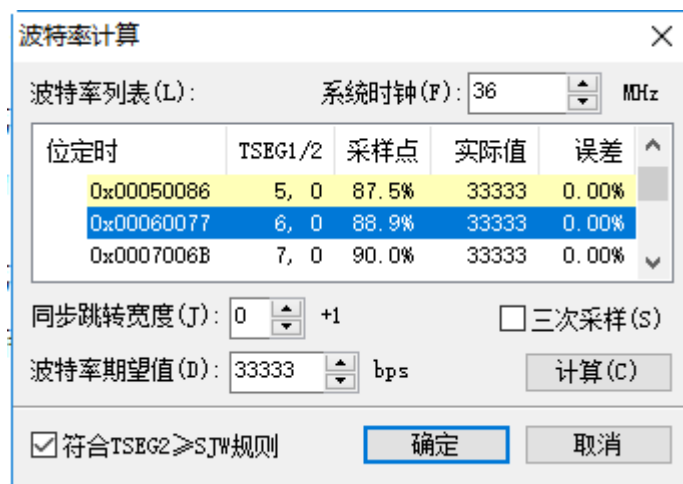


Figure 1 CAN baud rate calculation software

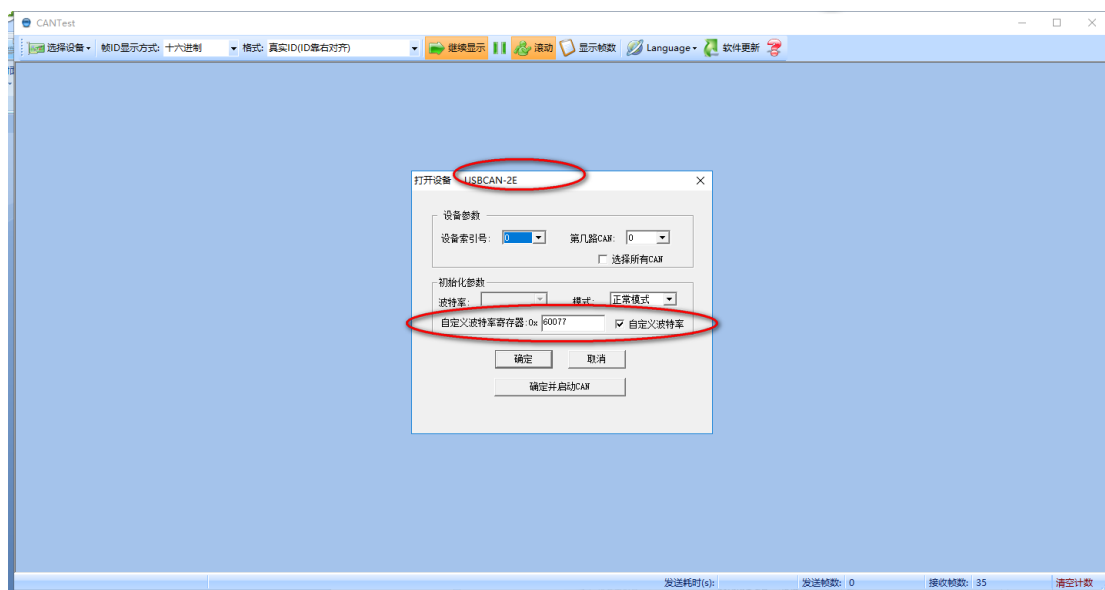
This tool is simple and convenient, you can follow the steps below to realize the baud rate calculation easily.

- Fill in the system clock of CAN controller in the first box (system clock) on the top of the software. The products, crystal oscillator is 36Mhz, needing to fill in "36000" (Model: USBCAN-2E-U);
- In the second box (baud rate expectation) under the software, fill in the actual baud rate that the user needs. For example, if you need a baud rate of 500Kbps, you need to fill in "500000"; if you need a baud rate of 33.333Kbps, you need to fill in "33333".
- By default, checks the box of "Comply with TSEG2≥SJW rule";
- In the lower right corner of the software, you can check "three times sampling". Sampling rate can be selected to sample once or sample three times, normally, in the case of low baud rate (less than 40Kbps) to use three samples, which can effectively eliminate bus burrs. At high baud rates, one sample is usually used;
- Once the baud rate requirements above are filled in, click on the Calculate button

(Calculate) and the baud rate used by the user will be calculated in the table on the left.

The values listed in the table all meet the requirements according to the CAN protocol, and the user can choose any set of data as the current baud rate value. However, if there is no data calculated in the table after clicking the Calculate button, it means that according to the crystal oscillator frequency set by the user and the required baud rate, the result can not be calculated to meet the requirements, in this case, the user can try to change the baud rate to recalculate.

Open the CANtest or CANpro software of Zhou Ligong, click the "Device Operation" menu, select the model "USBCAN-2E-U", and then check the (Customized Baud Rate) option in the "Parameter Confirmation" dialog box. In the "Parameter Confirmation" dialog box, check the "Custom Baud Rate" option. Fill in the 32-bit value (hexadecimal) of "Bit Timing" calculated by Baud Rate Calculator into the box. Configure the baud rate of 33.333k as shown in the figure above.



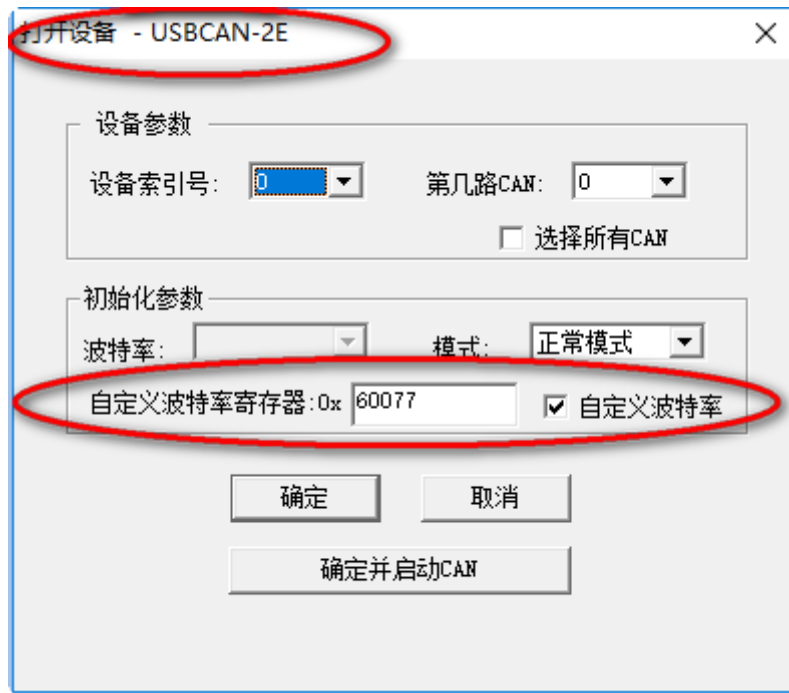


Figure 3 Baud rate configuration

Note: This software is only suitable for calculating the baud rate of our products, e.g. USBCAN adapter/CANalyst-II. The data calculated by this software may not be applicable to CAN controllers of other companies.

2. Schedule: Commonly Baud Rate and Bus Timing Register Settings Comparison Table

CAN Baud Rate	Timing0(BTR0)
10 Kbps	0x1C00E0
20 Kbps	0x1600B3
40 Kbps	0x1C0037
50 Kbps	0x1C002C
80 Kbps	0x60031
100 Kbps	0x160023
125 Kbps	0x1C0011
200 Kbps	0x60013
250 Kbps	0x1C0008
400 Kbps	0x60009
500 Kbps	0x60007
666 Kbps	0x60005
800 Kbps	0x60004
1000 Kbps	0x60003
33.33 Kbps	0x60077
66.66 Kbps	0x6003B
83.33 Kbps	0x6002F